

Course class documentation

A $\LaTeX$ class for writing courses

Felix RONDEAU

```
\title{course.cls}
\subtitle{A \LaTeX class for writing courses}
\author{Felix RONDEAU}
\maketitle
```

Section 1 Description

Part 1.1 Class features

1.1.1 Provided environments

```
\section{Description}
\subsection{Class features}
\subsubsection{Provided environments}
```

Definition 1

```
\begin{definition}
...
\end{definition}
```

Theorem 1

```
\begin{theorem}
...
\end{theorem}
```

Proof

```
\begin{proof}
...
\end{proof}
Ended by a QED  $\rightarrow$ 
```

□

Lemma 1

```
\begin{lemma}
...
\end{lemma}
```

Corollary 1

```
\begin{corollary}
...
\end{corollary}
```

Example 1

```
\begin{example}
...
\end{example}
```

Remark

```
\begin{remark}
...
\end{remark}
```

1.1.2 Aliases

The class provides aliases for commonly used mathematical objects. Here are some examples.
Mathematical sets

$$\mathcal{P} \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Code

```
\P \subset \N \subset \Z \subset \Q \subset \R \subset \C
```

Complex numbers and vectors

$$\Re\left(\overline{(b-a)} \times (d-c)\right) = \overrightarrow{AB} \cdot \overrightarrow{CD} = \left\| \overrightarrow{AB} \right\| \times \left\| \overrightarrow{CD} \right\| \times \cos\left(\widehat{\overrightarrow{AB}, \overrightarrow{CD}}\right)$$

Code $\left| \begin{array}{l} \backslash \text{Re}\text{Bigl}(\backslash \text{conj}\{(b-a)\} \backslash \text{times} (d-c)\text{Bigr}) = \backslash \text{vec}\{AB\} \backslash \text{cdot} \backslash \text{vec}\{CD\} = \backslash \text{norm}\{\backslash \text{vec}\{AB\}\} \\ \backslash \text{times} \backslash \text{norm}\{\backslash \text{vec}\{CD\}\} \backslash \text{times} \backslash \text{cos}\text{Bigl}(\backslash \text{widehat}\{\backslash \text{vec}\{AB\}, \backslash \text{vec}\{CD\}\}\text{Bigr}) \end{array} \right.$

Matrices

$$\forall M \in \mathcal{M}_n, M \in \text{GL}_n(\mathbb{K}) \iff \det M \neq 0$$

Code $\left| \backslash \text{forall} M \backslash \text{in} \mathcal{M}_{\{n\}}, M \backslash \text{in} \text{GL}_{\{n\}}(\mathbb{K}) \backslash \text{iff} \backslash \text{det} M \backslash \text{neq} 0 \right.$

Trigonometry, with definition of Arccos ($\backslash \text{acos}$), Arcsin ($\backslash \text{asin}$) and Arctan ($\backslash \text{atan}$), as well as hyperbolic functions like ch ($\backslash \text{ch}$), sh ($\backslash \text{sh}$) and th ($\backslash \text{th}$),

Arithmetics ($\backslash \text{div}$ and $\backslash \text{ncgm}$ are also available),

$$\forall p \in \mathcal{P}, \forall a \in \llbracket 1, p \rrbracket, p \nmid a \implies a^{p-1} \equiv 1 [p]$$

Code $\left| \backslash \text{forall} p \backslash \text{in} \mathcal{P}, \backslash \text{forall} a \backslash \text{in} \backslash \text{iset}\{1, p\}, p \backslash \text{ndiv} a \backslash \text{implies} a^{\{p-1\}} \backslash \text{cgm}\{1\}\{p\} \right.$

Limits and regularity classes,

$$\exists f \in \mathcal{F}(\mathbb{R}) \cap \mathcal{C}(\mathbb{R}) \cap \mathcal{D}(\mathbb{R}) : f(x) \xrightarrow{x \rightarrow 0} 0$$

Code $\left| \backslash \text{exists} f \backslash \text{in} \backslash \text{Func}(\mathbb{R}) \backslash \text{cap} \backslash \text{Cont}(\mathbb{R}) \backslash \text{cap} \backslash \text{Diff}(\mathbb{R}) : f(x) \backslash \text{arrowlim}\{x \backslash \text{to} 0\} 0 \right.$

Probabilities, with $\mathbb{P}$ ($\backslash \text{prob}$), $\mathbb{E}$ ($\backslash \text{expect}$) and $\mathbb{V}$ ($\backslash \text{variance}$), comprehension sets :

$$[a, b[= \{x \in \mathbb{R} \mid a \leq x < b\}$$

Code $\left| [a, b[= \backslash \text{set}\{x \backslash \text{in} \mathbb{R} \backslash \text{where} a \backslash \text{leq} x < b\} \right.$

...and many more to come !

Section 2 implementation

```

Code %% course.cls
%% Copyright 2024 RONDEAU Felix
%
% This work may be distributed and/or modified under the
% conditions of the LaTeX Project Public License, either version 1.3
% of this license or (at your option) any later version.
% The latest version of this license is in
%   https://www.latex-project.org/lppl.txt
% and version 1.3c or later is part of all distributions of LaTeX
% version 2024 or later.
%
% This work has the LPPL maintenance status 'maintained'.
%
% The Current Maintainer of this work is RONDEAU Felix.
%
% This work consists of the files course.cls and course-fr.cls

%-----
%          CLASS CONFIGURATION
%-----

\NeedsTeXFormat{LaTeX2e}
\ProvidesClass{course}[A LaTeX class for typesetting courses.]

\LoadClass{article}

%-----
%          PACKAGES AND OTHER DOCUMENT CONFIGURATIONS
%-----

\usepackage{amsmath, amsfonts, mathtools, amsthm, thmtools, amssymb}
\usepackage{bm, cancel, nicefrac}
\usepackage{dvipsnames}{xcolor}
\usepackage[many]{tcolorbox}
\usetikzlibrary{decorations.pathmorphing}
\usepackage[b]{esvect}
\usepackage{siunitx}
\usepackage{titlesec} % Title formatting
\usepackage{enumitem}
\usepackage{array}
\usepackage{stmaryrd}

%-----
%          PAGE LAYOUT
%-----

\usepackage{geometry}

\geometry{
  paper=a4paper,
  top=1.75cm,
  bottom=1.75cm,
  left=3.5cm,
  right=2cm,
  headheight=0.75cm,
  footskip=0.5cm, % Space from the bottom margin to the baseline of the footer
  headsep=0.5cm, % Space from the top margin to the baseline of the header
  marginparwidth=.2\paperwidth,
  %showframe, % Uncomment to show how the type block is set on the page
}

% Header and Footer setup
\def\ttitle#1{\gdef\ttitle{#1}}
\def\subtitle#1{\gdef\subtitle{#1}}
\def\author#1{\gdef\author{#1}}
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{}
\fancyhf[lh]{\ttitle}
\fancyhf[rh]{\itshape\nouppercase{\leftmark}}
\fancyhf[rf]{\thepage}

```

```

%-----
%          FONTS
%-----

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}

\usepackage{raleway}
\usepackage{libertine}

% For \Re and \Im
\usepackage[scr=boondox, % heavily sloped
             cal=cm] % slightly sloped
{mathalpha}

% Set Linux Libertine as the default roman font
\renewcommand{\rmdefault}{LinuxLibertineT-LF}

% Set Raleway as the default sans-serif font
\renewcommand{\sfdefault}{Raleway-TLF}

%-----
% CAPTION
%-----

\renewcommand\maketitle{
  \thispagestyle{empty}
  \begin{center}
    \fbox{\bfseries\sffamily\LARGE\parbox{\dimexpr\linewidth-2\fboxsep-2\
fboxrule\relax}{\centering\@title}}
  \end{center}
  \vspace{-8pt}
  {\bfseries\itshape\sffamily\footnotesize \@subtitle\hfill\@author}
  \if\relax\@subtitle\relax
    \if\relax\@another\relax
      \vspace{-\baselineskip}
    \fi
  \fi
  \fi
  \\
}

%-----
% USEFUL MATHS COMMANDS
%-----

% Common sets
\newcommand{\N}{\ensuremath{\mathbb{N}}}
\newcommand{\Z}{\ensuremath{\mathbb{Z}}}
\newcommand{\Q}{\ensuremath{\mathbb{Q}}}
\newcommand{\R}{\ensuremath{\mathbb{R}}}
\newcommand{\C}{\ensuremath{\mathbb{C}}}
\newcommand{\U}{\ensuremath{\mathbb{U}}}
\newcommand{\K}{\ensuremath{\mathbb{K}}}
\renewcommand{\P}{\ensuremath{\mathcal{P}}}
\newcommand{\M}{\ensuremath{\mathcal{M}}}
\newcommand{\GL}{\ensuremath{\mathrm{GL}}}

% Complex numbers
\let\conj\overline
\renewcommand{\Re}{\mathscr{R}\!e}
\renewcommand{\Im}{\mathscr{I}\!m}

% Functions set and regularity classes
\newcommand{\Func}{\mathcal{F}}
\newcommand{\Cont}{\mathcal{C}}
\newcommand{\Diff}{\mathcal{D}}

% Trigonometry
\DeclareMathOperator{\acos}{Arccos}
\DeclareMathOperator{\asin}{Arcsin}
\DeclareMathOperator{\atan}{Arctan}
\DeclareMathOperator{\ch}{ch}
\DeclareMathOperator{\sh}{sh}
\let\th\undefined
\DeclareMathOperator{\th}{th}

% Arithmetic

```

```

\renewcommand{\div}{\mid}
\newcommand{\ndiv}{\nmid}
\newcommand{\cgm}[2]{\equiv #1 \left[#2\right]} % ConGruent Modulo
\newcommand{\ncgm}[2]{\not\equiv #1 \left[#2\right]}
\newcommand{\iset}[1]{\left\llbracket #1 \right\rrbracket}

% Sets
\newcommand{\set}[1]{\{\,\,#1\,\}}
\newcommand{\where}{\mid}

% Vectors
\renewcommand{\vec}{\vv}
% TODO: Change font
\newcommand{\norm}[1]{\left\lVert#1\right\rVert}

% Limits
\newcommand{\arrowlim}[1]{\xrightarrow[#1]{}}

% Differentials
\newcommand{\dd}{\mathrm d}
%
% Probabilities
\newcommand{\prob}{\mathbb P}
\newcommand{\expect}{\mathbb E}
\newcommand{\variance}{\mathbb V}

% Other
\let\leq\leqslant
\let\geq\geqslant
\newcommand{\prop}{\mathcal P}

%-----
% SECTION / SUBSECTION STYLE
%-----

\newlength{\spacelength}
\settowidth{\spacelength}{\normalfont\ }

\titleformat{\section}[block]
{\bfseries\sffamily\Large}
{}
{0pt}
{\makebox[0pt][r]{Section \thesection\hspace{\marginparsep-2\spacelength}}}

\titleformat{\subsection}[block]
{\bfseries\sffamily\large}
{}
{0pt}
{\makebox[0pt][r]{Part \thesubsection\hspace{\marginparsep-2\spacelength}}}

\titleformat{\subsubsection}[block]
{\bfseries\sffamily}
{}
{0pt}
{\makebox[0pt][r]{\thesubsubsection\hspace{\marginparsep-2\spacelength}}}

%-----
% LISTS STYLE
%-----

% Enumerate without top margin
\newlist{enumerate-}{enumerate}{3}
\setlist[enumerate-]{nosep, itemsep=4pt, leftmargin=*, label=\arabic*}

% Itemize without top margin
\newlist{itemize-}{itemize}{3}
\setlist[itemize-]{label=\textbullet, nosep, itemsep=4pt, leftmargin=*}

%-----
% THEOREM STYLE
%-----

\declaretheoremstyle[
  headfont={\bfseries\sffamily\footnotesize},
  notefont={\bfseries\sffamily\footnotesize},
  bodyfont={\normalfont},

```

```

        headpunct={\relax},%\newline,
        notebraces={}{},
        headformat={%
            \makebox[0pt][r]{\ifthenelse{\equal{\NOTE}{} }\{NAME\}{\NOTE}\ \
NUMBER\hspace{\marginparsep}}\hspace{-\spacelength}},
    ]{numbered}
\declaretheoremstyle[
    headfont={\bfseries\sffamily\footnotesize},
    notefont={\bfseries\sffamily\footnotesize},
    bodyfont={\normalfont},
    headpunct={\relax},%\newline,
    notebraces={}{},
    headformat={%
        \makebox[0pt][r]{\ifthenelse{\equal{\NOTE}{} }\{NAME\}{\NOTE}\ \
hspace{\marginparsep}}\hspace{-\spacelength}},
    ]{unnumbered}
\declaretheoremstyle[
    headfont={\footnotesize\itshape},
    notefont={\normalfont},
    bodyfont={\normalfont},
    headpunct={\relax},
    headformat={%
        \makebox[0pt][r]{\ifthenelse{\equal{\NOTE}{} }\{NAME\}{\NOTE}\ \
hspace{\marginparsep}}\hspace{-\spacelength}},
    ]{claim}

\theoremstyle{unnumbered}
\newtheorem{blank}{}

% Theorem
\declaretheorem[name=Theorem, style=numbered]{theorem}
\tcolorboxenvironment{theorem}{
    boxrule=0pt,
    boxsep=0pt,
    enhanced jigsaw,
    opacityback=0,
    colframe=black,borderline={1pt}{0mm}{black},
    sharp corners,
    before skip=8pt,
    after skip=8pt,
    left=5pt,
    right=5pt,
    breakable,
}

\newenvironment{theorems}
{\begin{theorem}[Theorems]}\end{theorem}}

\newenvironment{theoremlist}
{\begin{theorems}\begin{enumerate-}}{\end{enumerate-}\end{theorems}}

% Proof and building
\let\proof\relax
\declaretheorem[
    name=Proof,
    style=unnumbered,
    qed=\qedsymbol]{proof}

\tcolorboxenvironment{proof}{
    boxrule=0pt,
    boxsep=0pt,
    blanker,
    borderline west={0.5pt}{0pt}{black!60!white},
    before skip=8pt,
    after skip=8pt,
    left=5pt,
    right=5pt,
    breakable,
}

\newenvironment{building}
{\begin{proof}[Building]}\end{proof}}

\newenvironment{proofs}
{\begin{proof}[Proofs]}\end{proof}}

```

```

\newenvironment{prooflist}
{\begin{proofs}\begin{enumerate-}}{\end{enumerate-}\end{proofs}}

% Definition
\theoremstyle{numbered}
\newtheorem{definition}{Definition}

\newenvironment{definitions}
{\begin{definition}[Definitions]]{\end{definition}}

\newenvironment{deflist}
{\begin{definitions}\begin{itemize}[nosep, itemsep=4pt, leftmargin=*]]{\end{itemize}\end{
definitions}}

\tcolorboxenvironment{definition}{
    boxrule=0pt,
    boxsep=0pt,
    enhanced jigsaw,
    opacityback=0,
    colframe=black,borderline={1pt}{0mm}{black,dashed},
    sharp corners,
    before skip=8pt,
    after skip=8pt,
    left=5pt,
    right=5pt,
    breakable,
}

% Properties
\theoremstyle{numbered}
\newtheorem{property}{Property}

\newenvironment{properties}
{\begin{property}[Properties]]{\end{property}}

\newenvironment{proplist}
{\begin{property}\begin{itemize}[nosep, itemsep=4pt, leftmargin=*]]{\end{itemize}\end{
property}}

\tcolorboxenvironment{properties}{
    boxrule=0pt,
    boxsep=0pt,
    enhanced jigsaw,
    opacityback=0,
    colframe=black,borderline={1pt}{0mm}{black},
    sharp corners,
    before skip=8pt,
    after skip=8pt,
    left=5pt,
    right=5pt,
    breakable,
}

% Lemma
\theoremstyle{numbered}
\newtheorem{lemma}{Lemma}
\tcolorboxenvironment{lemma}{
    boxrule=0pt,
    boxsep=0pt,
    blanker,
    borderline west={0.8pt}{0pt}{decoration={snake,aspect=0.3,segment length=3mm,
amplitude=1pt},decorate},
    before skip=8pt,
    after skip=8pt,
    sharp corners,
    left=5pt,
    right=5pt,
    breakable,
}

% Corollary
\theoremstyle{numbered}
\newtheorem{corollary}{Corollary}
\tcolorboxenvironment{corollary}{
    boxrule=0pt,
    boxsep=0pt,
    blanker,

```

```

        borderline west={0.8pt}{0pt}{decoration={zigzag,amplitude=1pt,pre=moveto,segment
length=5pt,angle=42},decorate},
        before skip=8pt,
        after skip=8pt,
        sharp corners,
        left=5pt,
        right=5pt,
        breakable,
    }

% Remarks
\theoremstyle{unnumbered}
\newtheorem{remark}{Remark}

\newenvironment{remarks}
{\begin{remark}[Remarks]}\end{remark}}

\newenvironment{remarklist}
{\begin{remarks}\begin{itemize}[nosep, itemsep=4pt, leftmargin=*]}\end{itemize}\end{
remarks}}

\tcolorboxenvironment{remark}{
    boxrule=0pt,
    boxsep=0pt,
    blanker,
    borderline west={1pt}{0pt}{black,dotted},
    before skip=8pt,
    after skip=8pt,
    sharp corners,
    left=5pt,
    right=5pt,
    breakable,
}

% Example
\theoremstyle{numbered}
\newtheorem{example}{Example}

\newenvironment{examples}
{\begin{example}[Examples]}\end{example}}

\newenvironment{exlist}
{\begin{examples}\begin{itemize}[nosep, itemsep=4pt, leftmargin=*]}\end{itemize}\end{
examples}}

\tcolorboxenvironment{example}{
    boxrule=0pt,
    boxsep=0pt,
    blanker,
    borderline west={1pt}{0pt}{black},
    sharp corners,
    before skip=8pt,
    after skip=8pt,
    left=5pt,
    right=5pt,
    breakable,
}

```